

scenarios. We next propose algorithms that use these properties to solve the power-minimization optimization problem.

4 Application Placement Algorithms

In this section, we describe the various placement algorithms designed for minimizing the overall cost, while meeting a fixed performance SLA. The algorithms assume that a performance manager provides them with a VM size for each application, that can meet its performance goals. We start with a brief description of the key ideas behind the algorithms.

4.1 Algorithm Idea

Our application placement algorithms that minimize the power and migration costs are based on three key observations

1. Estimating the power cost of a given configuration may not be possible because power drawn by a server depends on the exact application mix on a server.
2. Background Independence Property and Ordering Property allows one to pick a server for an application that will minimize the incremental power due to the new application. Hence, local searches are feasible.
3. The above properties may not hold always if servers are equally loaded. However, the properties will definitely hold if we compare an idle server with a loaded server.

The first two observations dictate the design of algorithms to be based on local searches. One can view the application placement problem as a bin-packing problem with differently sized bins. The servers represent the bins and the virtual machines represent the balls. The power drawn by a server is represented as the cost of the bin and the power-minimizing allocation is a packing that minimizes the cost of the packed bins. Since power drawn by a server depends on the actual mix of applications, the cost of packing a bin varies depending on the balls being packed in the bin. Bin packing has many local search algorithms like First-Fit Decreasing(FFD) [25], Best-Fit and Worst-fit.

We took a close look at FFD, where balls are ordered by size (largest first). The balls are then packed in the first bin that can accommodate them. We observed that if servers are ordered based on power efficiency (static power per unit capacity), then the FFD algorithm can employ the Ordering property to minimize power. This is because FFD unbalances load, and as per our third observation, the Ordering property always holds if we are comparing an idle server with a loaded server. Further, by placing servers based on their power efficiency, we ensure that more power efficient servers are the ones that are utilized first. Finally, FFD has good theoretical bounds and is also known to perform well in practice [25]. Hence, we focus on adapting First Fit to work for different sized bins with ball-dependent cost functions.